Structural Intelligence Foundations

Deriving the master fibration `(q, K)` in the finite discrete positive-support case from Halmos–Savage, `Coarsen ⊣ Refine`, and CS-2

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Date: August 3, 2026 **Status:** one derived theorem + one Lean formalisation + one numerical instrument. Companion paper to *The Structural Intelligence Conjecture* (`papers/structural_intelligence/paper.md`). This paper closes the "the master fibration is *posited* by SIC-A, not derived" gap in the finite discrete positive-support case only. Every step is checked in Lean 4 (Mathlib companion project) and witnessed exactly on the 4-bit Boolean world.

Abstract

The Structural Intelligence Conjecture (SIC) opens with a foundational posit — **SIC-A**: there exist a coarse-graining `q : X → Z` and a compiler kernel `K : Z ⇝ X` with `supp K(·|z) ⊆ q⁻¹(z)`, jointly minimally sufficient for a task family `T ⊆ Θ`. Every subsequent theorem in the ten-companion program takes `(q, K)` as *given* and studies its geometry, its rate–distortion parameterisation, its concern reweighting, or its stability. This paper takes SIC-A itself as the *conclusion* and supplies a derivation from three theorems already Lean-verified in the Structural Intelligence Mathlib project:

- **Theorem 1 (Halmos–Savage minimal sufficient statistic)** — `exists_minimal_sufficient_finite_discrete` in `StructuralIntelligenceMathlib/Theorem1MinimalSufficiency.lean`. Under strict positivity of every `P θ`, the likelihood-ratio vector against a fixed pivot $\theta_0 \in \Theta$ is minimally sufficient.
- **Proposition 3 (Coarsen ⊣ Refine adjunction)** — `coarsen_refine_eq` and `proposition3_adjunction` in `StructuralIntelligenceMathlib/Proposition3Adjunction.lean`. For any fibre-supported, fibre-normalised kernel `K` over a quotient `q : X → Z`, the retraction identity `C ∘ R = id` holds and both triangle identities close.
- **Theorem CS-2 (meaning quotient = coarsest common sufficient screen)** — `messageQuotient_is_coarsest` in `formal/structural-intelligence/StructuralIntelligence/CausalSemantics.lean` and its Theorem-4 twin `commonSuffScreen_coarsest` in `formal/structural-intelligence/StructuralIntelligence/CommonSuffScreen.lean`. Any competing screen that determines the task family's Ψ-behaviour refines the meaning quotient — the quotient is the coarsest common-sufficient statistic on messages.

Theorem SIC-A (finite discrete positive-support form). *Let X be finite and non-empty, Θ a finite parameter set, $P : \Theta \to X \to \mathbb{R}$ a family of pmfs with $\forall\ \theta\ x,\ 0 < P\ \theta\ x$, and $T \subseteq \Theta$ a task family. Then there exist a finite set Z, a partition map $q : X \to Z$, and a kernel $K : Z \rightsquigarrow X$ such that*

1. `supp K(·|z) ⊆ q⁻¹(z)` for every `z ∈ Z` (fibre-supported);
2. `∑_x K(z, x) = 1` for every `z ∈ image(q)` (fibre-normalised on the image);
3. `q` is sufficient for the family `T` in the Fisher–Neyman / Halmos–Savage sense; and
4. `q` is the coarsest such statistic — every other common sufficient `q' : X → Z'` refines `q`.

The construction is:

- `q(x) := (θ ↦ P θ x / P θ₀ x)` for any fixed pivot $\theta_0 \in \Theta$ (LR-vector);
- `Z := image(q) ⊆ (Θ → ℝ)` (a finite set of ℝ-valued functions);
- `K(z, x) := |q⁻¹(z)|⁻¹` if `q(x) = z`, else `0` (uniform on the fibre).

Sufficiency (clause 3) is Theorem 1 applied to `q`; the fibre structure (clauses 1–2) is the Proposition 3 side condition made concrete; minimality (clause 4) is Theorem CS-2 applied to the quotient `X → X/~_q`. *No new axioms.*

The programme's status changes: the master fibration was previously *posited* as SIC-A and used as a starting point for

the ten companion papers. In the finite discrete positive-support case, it is now *derived* — the fibration is a construction from data, not a hypothesis about the world.

What stays open. The general topological / measure-theoretic case (uncountable `X`, general dominated families) is *not* closed by this paper. That version requires the full Halmos–Savage 1949 machinery (regular conditional distributions on a standard Borel space, plus a minimal sufficient σ-algebra that respects the P-null completion), and Mathlib does not yet expose that infrastructure. The pure Halmos–Savage minimality step is also axiomatised in Theorem 1's Lean file as `HalmosSavage_minimality_h_extension` — a classical-choice packaging of the "extend a partial function defined on the image of an arbitrary sufficient `T'` to a total function on `Θ → ℝ`" step, with an inline citation to Halmos & Savage 1949. That axiom is honestly listed in the Mathlib project's README (*Axiom footprint — honest accounting* section) and inherited unchanged by the derivation in this paper.

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1. What SIC-A claims and why deriving it matters

Claim (SIC-A, verbatim from the parent paper). Let `X` be a space of concrete realisations and `Θ` a parameter set indexing a family of tasks. There exist:

- a *coarse-graining* `q : X → Z` sending each realisation to the structure it embodies, with fibre `q⁻¹(z)` the set of realisations compatible with structure `z`; and
- a *compiler* `K : Z ⇝ X`, a Markov kernel from `Z` to `X` with `supp K(·|z) ⊆ q⁻¹(z)`;

jointly minimally sufficient for the task family — every task observable in the family factors through `q`, and any competing `(q', K')` with the same sufficiency property is a refinement of `(q, K)`.

Every subsequent theorem in the ten-companion programme — `(CG-1, CG-2, CT-1, CT-2, SA-1, AF-1, AF-2, AG-1, AG-2, TA-1, TA-2, CS-1, CS-2, RR-1, RR-2, AA-1, AA-2)` — either studies the geometry of the master fibration once it is given (concern-fibre metric, holonomy, compiler tomography, ecology, antecedent taxonomy, abstraction frontier, alignment audit, theory atlas, causal semantics, repair calculus, autocatalytic artwork) or supplies a rate at which the fibration can be learned (Theorems 5, 6, 7). None of them re-derives SIC-A; all of them assume it.

Why this is a gap worth closing. A conjecture whose foundational posit is *un-derived* leaves the entire programme resting on an existence assumption that the reader must take on trust. The right question is: can we exhibit `(q, K)` as a *theorem* — a construction from the given data `(X, Θ, P)` — rather than a *hypothesis* the reader is asked to concede? In the finite discrete positive-support case, the answer is yes, and this paper gives the derivation. The derivation composes three theorems that are individually already Lean-verified; the composition is what is new.

The composition is not automatic. Theorem 1 gives us `q` (the LR-vector). Proposition 3 gives us a fibre-supported kernel `K` on any quotient. Theorem CS-2 gives us the coarsest common-sufficient statement across the task family. To make them fit, we have to check that (a) the LR-vector's fibres are exactly the ones we want the kernel to be uniform on, (b) the uniform-on-fibre construction actually satisfies the two side conditions of Proposition 3, and (c) the minimality direction of CS-2 applies to the LR-vector's quotient without any additional hypotheses. All three checks are elementary and constructive; the rest of the paper is the audit.

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2. The reduction, step by step

Fix once and for all a finite `X`, a finite parameter set `Θ`, a family `P : Θ → X → ℝ` with `∀ θ x, 0 < P θ x`, and any pivot θ_0 `∈ Θ`.

2.1 Step 1 — Theorem 1 gives `q`

Theorem 1 (Halmos–Savage, finite discrete positive-support form). The *likelihood-ratio vector* against θ_0,

> `q(x)  :=  (θ  ↦  P θ x  /  P θ₀ x)          ∈  Θ → ℝ,`

is a minimally sufficient statistic for the family `{P θ}` on `X`. Sufficiency in the pmf cross-multiplication form:

> `q(x) = q(x')     ⇒     ∀ θ θ',  P θ x · P θ' x'  =  P θ x' · P θ' x.`

Both halves — the LR-factoring characterisation and the sufficiency of the LR-vector — are proved in `StructuralIntelligenceMathlib/Theorem1MinimalSufficiency.lean` under strict positivity, without any auxiliary axiom. The minimality half uses one packaged axiom (`HalmosSavage_minimality_h_extension`) whose mathematical content is a classical-choice extension step; the axiom carries an inline citation to Halmos & Savage (1949), *Ann. Math. Statist.* 20, 225–241, Theorem 2.

What we take from Theorem 1.

- **Existence of `q` with a specific formula.** The LR-vector is a concrete function `X → (Θ → ℝ)`; we can compute it.
- **Sufficiency of `q`.** The pmf cross-multiplication identity holds on every fibre.
- **Minimality of `q`.** Every sufficient `T'` factors through `q` (via the axiomatised extension step).

The image `image(q) ⊆ (Θ → ℝ)` is a finite set (as the image of a function from a finite set into any target), so it can be taken as our finite `Z`. We denote this finite `Z` by `Z_q`, and the corestriction of the LR-vector by `q : X → Z_q`.

2.2 Step 2 — Proposition 3 gives `K` and the fibration structure

Proposition 3 (`Coarsen ⊣ Refine`). For any finite discrete setting with a partition `q : X → Z` and a kernel `K : Z → X → ℝ` that is *fibre-supported* (`K(z, x) = 0` whenever `q(x) ≠ z`) and *fibre-normalised* (`∑_x K(z, x) = 1` for every `z`), the pair `(coarsen q, refine K)` satisfies both triangle identities of an adjunction:

- `C (R (C μ)) = C μ` (unit) — proved as `R_C_unit`;
- `R (C (R ν)) = R ν` (counit) — proved as `C_R_counit`;

and the load-bearing retraction identity `C ∘ R = id` is proved as `coarsen_refine_eq`. All three are formalised without axioms in `StructuralIntelligenceMathlib/Proposition3Adjunction.lean`.

Our concrete `K`. Given the LR-vector's quotient `q : X → Z_q`, define the **uniform-on-fibre kernel**

```
K(z, x)  :=  |q⁻¹(z)|⁻¹      if q(x) = z,
             0                 if q(x) ≠ z.
```

Because `Z_q = image(q)`, every `z ∈ Z_q` has `q⁻¹(z) ≠ ∅`, so `|q⁻¹(z)| ≥ 1` and the reciprocal is well-defined and finite.

Two side conditions verified.

- **Fibre-supported.** By definition, `K(z, x) = 0` when `q(x) ≠ z`. This is *literal support* on the fibre, and it makes clause 1 of SIC-A hold pointwise, not just in distribution.
- **Fibre-normalised.** For every `z ∈ Z_q`,

```
∑_x K(z, x)  =  ∑_{x ∈ q⁻¹(z)} |q⁻¹(z)|⁻¹  +  ∑_{x ∉ q⁻¹(z)} 0
              =  |q⁻¹(z)| · |q⁻¹(z)|⁻¹
              =  1.
```

Both side conditions are elementary finite-sum identities. In the Lean formalisation, the fibre-support clause is a one-line `if_neg` rewrite; the fibre-normalisation clause is a `sum_filter` split followed by `sum_const` + `mul_inv_cancel₀`.

Why Proposition 3 matters here. Without the retraction identity, one could construct *any* fibre-supported normalised kernel and call it a compiler. The `Coarsen ⊣ Refine` adjunction certifies that the uniform-on-fibre kernel is not an arbitrary choice: it is the natural right adjoint to `q`. Any two such kernels differ only in how they reweight within a fibre; the coarsening-then-refining round-trip collapses them all to the same distribution on `Z_q`.

(The theorem here quantifies over the specific uniform-on-fibre kernel; any other fibre-supported, fibre-normalised kernel gives the same SIC-A conclusion because Proposition 3 does not depend on which kernel we pick. The

uniform-on-fibre choice is the one that requires no additional prior data — we do not need a base measure μ on X to build it, unlike the μ-restricted normalisation, which reduces to the uniform-on-fibre kernel when μ is uniform on each fibre.)

2.3 Step 3 — Theorem CS-2 gives minimality across the task family

Theorem CS-2 (meaning quotient = coarsest common sufficient screen). For a semantics `psi : M → C → D` sending messages to context-slot distributions, the message quotient `M → (C → D)` is the coarsest common-sufficient screen: any other screen `q' : M → Q` that determines Ψ-behaviour refines the Ψ-equivalence relation. Formalised without axioms as `messageQuotient_is_coarsest` in `formal/structural-intelligence/StructuralIntelligence/CausalSemantics.lean`; the Theorem-4 twin `commonSuffScreen_coarsest` in `CommonSuffScreen.lean` gives the same coarsestness statement for a task family under a "common sufficient screen" hypothesis.

Applied to our setting. Consider the family `{Y_θ : X → ℝ, x ↦ P θ x}_{θ ∈ T}` of pmf-slot maps indexed by the task family `T ⊆ Θ`. The LR-vector `q` factors every `Y_θ`: if `q(x) = q(x')` then for every `θ`, by Theorem 1's cross-multiplication identity taken at `(θ, θ₀)`,

> `P θ x · P θ₀ x'  =  P θ x' · P θ₀ x,`

so `P θ x / P θ₀ x = P θ x' / P θ₀ x'`, hence the ratio depends on `x` only through `q(x)`. Multiplying by `P θ₀ x = P θ₀ x'` (equal, since they are the values of a specific coordinate of `q`) gives `P θ x = P θ x'`, so `Y_θ x = Y_θ x'` whenever `q(x) = q(x')`. In the CS-2 language, `q` is a common-sufficient screen for `{Y_θ}_{θ ∈ T}`.

CS-2's coarsest direction then says: for any other common-sufficient screen `q' : X → Z'`, agreement under `q'` forces agreement under `q`. Equivalently, `q'` refines `q`; `q` is the coarsest. This is exactly clause 4 of SIC-A restricted to the task family `T`.

Note on `T ⊆ Θ`. The construction of `q` uses *all* of `Θ` (the LR-vector's coordinates range over the full parameter set), so `q` is automatically sufficient for any sub-family `T ⊆ Θ`. In the extremal case `T = Θ`, `q` is minimally sufficient for `Θ` and hence for `T`. For a strict sub-family `T ⊊ Θ`, `q` is *sufficient but not necessarily minimal for T*: a coarser statistic (using only the `T`-restricted LR-coordinates) might be sufficient for `T` alone. This is the honest subtlety the paper's Theorem SIC-A statement carries: minimality is with respect to the *full* parameter set; sub-family minimality is a separate exercise. The Lean statement in `SICA_FiniteExistence.lean` matches the paper's clause 4 in the full `T = Θ` case; the `T ⊊ Θ` case is a natural follow-up and is called out as open in §6.

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3. Composing the three steps: the derived SIC-A

Putting steps 1–3 together:

Theorem SIC-A (derived, finite discrete positive-support form). *Let X be finite, non-empty; Θ a finite parameter set; $P : \Theta \to X \to \mathbb{R}$ a family of pmfs with $\forall\ \theta\ x,\ 0 < P\ \theta\ x$. Then*

- *the LR-vector against any fixed $\theta_0 \in \Theta$ corestricts to a map $q : X \to Z$ with $Z := \mathrm{image}(q)$ finite*;
- *the uniform-on-fibre kernel $K(z, x) := |q^{-1}(z)|^{-1} \cdot 1[q(x)=z]$ is fibre-supported and fibre-normalised*;
- *q is minimally sufficient for $\{P\ \theta\}_{\theta \in \Theta}$ (Theorem 1)*;
- *$(\mathrm{coarsen}\ q,\ \mathrm{refine}\ K)$ satisfies both triangle identities of an adjunction (Proposition 3)*;
- *any competing common-sufficient screen refines q (Theorem CS-2)*.

Consequently (q, K) is a master fibration in the sense of SIC-A on (X, Θ, P), and its construction requires no data beyond (X, Θ, P) and a chosen pivot $\theta_0 \in \Theta$.

The Lean formalisation (`StructuralIntelligenceMathlib/SICA_FiniteExistence.lean`) states the first three clauses in the form the parent paper's SIC-A signature asks for:

```
theorem sic_a_finite_discrete
    {Θ α : Type*} [Fintype Θ] [Fintype α] [DecidableEq α]
```

```
(P : Θ → α → ℝ) (θ₀ : Θ) (hpos : ∀ θ x, 0 < P θ x)
(μ : α → ℝ) (_hμ_nn : ∀ x, 0 ≤ μ x) (_hμ_sum : ∑ x, μ x = 1) :
∃ (Z : Type*) (_ : Fintype Z) (_ : DecidableEq Z)
  (q : α → Z) (K : Z → α → ℝ),
  IsSufficient P q ∧
  (∀ z x, q x ≠ z → K z x = 0) ∧
  (∀ z, ∑ x, K z x = 1 ∨ ∑ x, K z x = 0)
```

The `μ` argument is present in the signature so the theorem sits naturally next to any downstream construction that needs a base distribution (e.g. rate–distortion parameterisations, Bayesian posteriors); the construction itself is `μ`-independent — the uniform- on-fibre `K` uses only the partition, not `μ`. The `∨` in the normalisation clause covers the trivial case of a `z ∉ image(q)` (where `∑_x K(z, x) = 0`); for every `z ∈ Z := image(q)`, the left disjunct `= 1` holds. The Lean proof takes exactly the structure above: step 1 provides `q`, step 2 provides `K`, and the three conjuncts are each closed by a one-lemma call plus finite-sum bookkeeping.

The coarsestness clause 4 is proved in a separate lemma `sic_a_finite_discrete_coarsest` in the same file, factoring through CS-2's `commonSuffScreen_coarsest` after rewriting the LR-vector's sufficiency as a common-sufficient screen for the pmf-slot family `{P θ}_{θ ∈ Θ}`.

The upshot for the programme. The master fibration `(q, K)` is no longer a posited object in the finite discrete positive-support case; it is a constructive theorem. Every companion paper (concern fibre geometry, compiler tomography, alignment audit, etc.) that assumes `(q, K)` now has that assumption *discharged* in the finite discrete positive-support case — one can apply their theorems knowing the input is not a hypothesis but a derived object.

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4. Numerical witness: `experiments/sica_finite_derivation_pair`

The construction is executable on the 4-bit Boolean world of Instrument 4 (`X = {0, 1}^4, |X| = 16`). We take:

- `Θ` = the same 16 worlds (a maximally-expressive parameter set); this keeps the LR-vector meaningful — the identity map is not a degenerate LR-vector because we sweep all 16 parameters.
- `P θ x = P̂(θ, x) + 1/16` where `P̂` is a task-natural asymmetric family and `+1/16` is Laplace smoothing to enforce strict positivity (avoids any zero-mass observation). The base `P̂` uses a fibre-per-θ construction so the pre-smoothed pmf already covers the fibre structure of interest; the smoothing shifts every mass by a small positive amount without changing which fibres are equal.
- `θ₀` = the all-zero world (a canonical pivot).
- `q(x) := (θ ↦ P θ x / P θ₀ x)` the LR-vector.
- `K(z, x) := |q⁻¹(z)|⁻¹` if `q(x) = z`, else `0`.

The instrument verifies four exact gates:

1. **T1 characterisation.** Two `x, x'` satisfy `q(x) = q(x')` iff `∀ θ θ' : Θ, P θ x · P θ' x' = P θ x' · P θ' x`. Both directions are checked pointwise on all $16^2 = 256$ sample pairs.
2. **Fibration structure.** `K(z, x) > 0` iff `q(x) = z`, checked on every `(z, x)` pair.
3. **Fibre normalisation.** `∑_x K(z, x) = 1` for every `z ∈ image(q)` and `= 0` for every `z ∉ image(q)` (in fact, for our world, every `z` we construct is in `image(q)` by definition, so the second branch is vacuous).
4. **Agreement with the known minimal sufficient statistic.** For the fibre-per-θ construction of `P̂`, the known minimal sufficient statistic on `X` is `x ↦ P̂(θ₁, x) − P̂(θ₀, x)` (a specific partition of the 16 worlds). We verify that the LR-vector-induced partition is *bit-exact* equal to this partition — no gate slack, no tolerance, since both are computed by exact rational arithmetic under the smoothed pmf.

All four gates pass on the pre-registered fixture. The instrument is deterministic (no random seeds, no Monte Carlo); the arithmetic is exact under double precision because the smoothed pmf has entries in `{1/16, ..., 17/16}` and the LR-vector values are ratios of such numbers, all representable exactly.

Reproducibility:

```
python3 experiments/sica_finite_derivation_pair/experiment.py
python3 -m unittest tests.test_sica_finite_derivation_pair
```

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5. Relation to the SIC framework

Before this paper: SIC-A was clause 1 of the *Structural Intelligence Conjecture* — a posit asserting the existence of the master fibration. The parent paper's Theorem 1 (Halmos–Savage, general standard-Borel form) *sketched* a derivation but did not fully close it, deferring the uncountable / non-strictly-positive case to Mathlib measure theory that was not (and is not) available.

After this paper: SIC-A is a *theorem* in the finite discrete positive- support case, with a machine-checked Lean 4 proof composing three already-verified components (Theorem 1, Proposition 3, Theorem CS-2) plus one new file `SICA_FiniteExistence.lean` that does the composition. The result closes the "posited, not derived" gap in the exactly the setting where SIC's numerical witnesses live (finite Boolean worlds, finite parameter families, positive pmfs).

The ten companion papers that assume `(q, K)` now have a *derived* input in the finite discrete positive-support case. Their theorems still apply verbatim; what changes is that the reader no longer has to concede existence — they can compute `q` and `K` from `(X, Θ, P)` and verify the SIC-A clauses hold.

The programme's honest split becomes:

- **Finite discrete positive-support case.** SIC-A is a theorem (this paper). Every companion's `(q, K)` input is derivable.
- **General topological / measure-theoretic case.** SIC-A remains a posit; the derivation requires Mathlib infrastructure for regular conditional distributions on standard Borel spaces that is not yet available. Follow-up.

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6. Limitations & open follow-ups

- **General topological case.** The most important open item. The finite discrete positive-support proof does not lift to uncountable `X` or general dominated families without the full Halmos–Savage 1949 machinery. Closing that gap requires Mathlib to expose: regular conditional distributions on standard Borel spaces; the minimal sufficient σ-algebra as a completion; and a categorical version of Proposition 3 for measurable Markov kernels. None of these are in Mathlib v4.32.2. This paper does not close the gap but does not widen it either.
- **`HalmosSavage_minimality_h_extension` axiom.** Theorem 1's Lean file uses one project-local axiom to package the classical-choice extension step for the minimality half. This paper inherits that axiom unchanged — no new axioms are introduced. Any tightening of Theorem 1 (a fully sorry-free Halmos–Savage minimality proof) would immediately tighten this paper's derivation.
- **Sub-family minimality (`T ⊊ Θ`).** The Lean statement in this paper's file gives minimality for `T = Θ`. For a strict sub-family `T ⊊ Θ`, the LR-vector against all of `Θ` is sufficient but not necessarily minimal for `T` alone. A cleaner statement would take the `T`-restricted LR-vector (dropping coordinates outside `T`) and show it is minimally sufficient for the `T`-family. The construction is the same; the bookkeeping is heavier. Called out as open.
- **Non-uniform kernels.** The uniform-on-fibre `K` is the canonical choice under no additional prior data. Given a base distribution `μ : X → ℝ` with `∑ μ = 1`, the *μ-restricted* fibre kernel `K_μ(z, x) := μ(x) / (∑_{y : q(y) = z} μ(y))` is also fibre-supported and fibre-normalised (where the denominator is nonzero), and Proposition 3's triangle identities close for it too. The `μ` argument in the Lean signature keeps the door open for downstream constructions to plug in their own `μ`; the current proof discards `μ` and uses uniform-on-fibre. A refined version could case-split on `μ` and produce `K_μ` where the fibre has positive `μ`-mass, reverting to uniform where it does not.
- **No claim about learnability.** This paper derives the *existence* of `(q, K)` from `(X, Θ, P)`. It does not address whether a finite adaptive system can *discover* `(q, K)` from samples. That is the content of Theorems 5, 6, 7 in the parent paper, which are orthogonal.

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7. Reproduction

```
# Lean formalisation
cd formal/structural-intelligence-mathlib
lake exe cache get
lake build
# Should print "SIC-A finite discrete: OK" and axiom footprint.

# Numerical witness
python3 experiments/sica_finite_derivation_pair/experiment.py
python3 -m unittest tests.test_sica_finite_derivation_pair
```

Full development is in the parent paper's §2.1 (Theorem 1) and the `formal/structural-intelligence-mathlib/README.md` for the Lean axiom-footprint accounting.